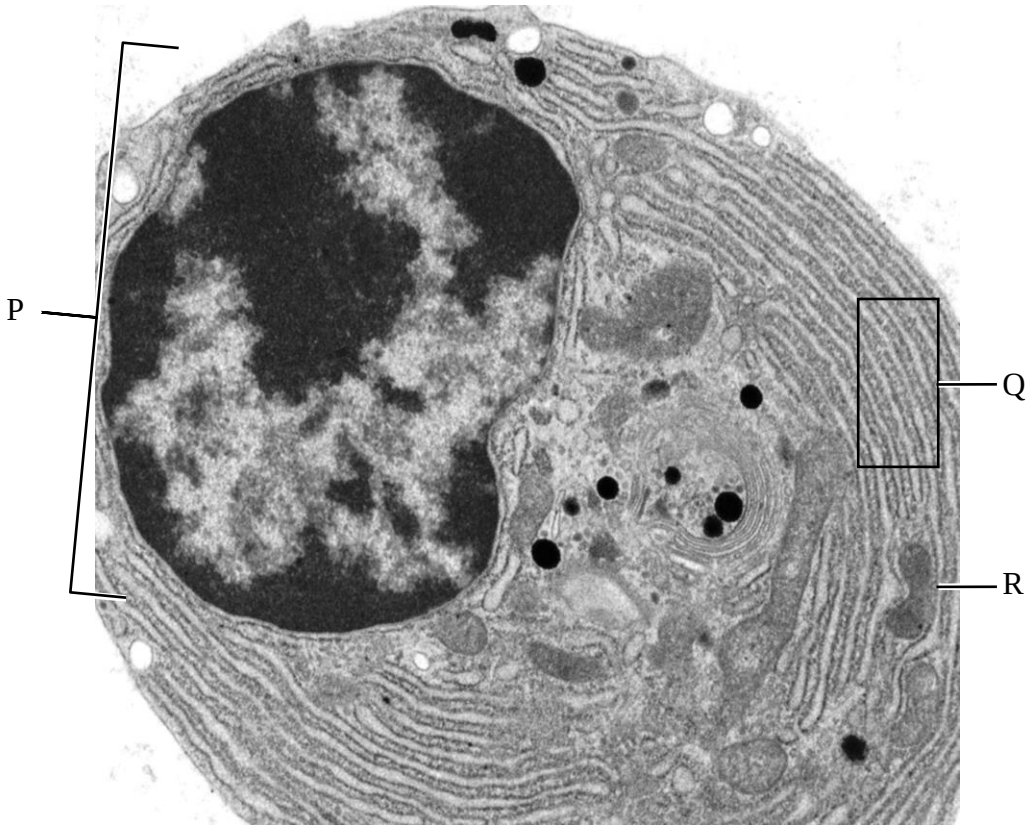
Short questions

|!|BCESA0270000001|!| ★★ **CRO**

The electron micrograph below shows part of a metabolically active cell.



(×4000)

(a) In which labelled structure does transcription occur? (1 mark)

(b) DNA cannot leave structure P. Name the molecule that carries information from P to the cytoplasm to direct translation. (1 mark)

(c) Identify the tiny particles on structure Q. What is their function? (2 marks)

(d) How does structure R contribute to the process of protein synthesis? (2 marks)

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##

(a) Structure P (1)

(b) Messenger RNA (1)

(c) The tiny particles are ribosomes. (1)

Ribosomes are the sites where polypeptides are synthesized. (1)

(d) Structure R is the site of ATP production. (1)

It supplies energy for protein synthesis. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCESA0270000002|!| ★★

Sickle-cell anaemia is a genetic disorder of the blood. It is caused by a mutation in the gene controlling the production of the β-chain of haemoglobin. Part of the base sequence on the template strand of DNA of the normal allele and the mutant allele for the β-chain are given below.

Normal allele: GGACTCCTC
Mutant allele: GGACACCTC

(a) Identify the type of mutation that has occurred. (1 mark)

_____

(b) Give the corresponding mRNA sequence for the part of the mutant allele. (1 mark)

_____

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(c) The table below shows the mRNA codons for some amino acids.

Amino acid	***mRNA codons***			
Glu	GAA	GAG		
Gly	GGU	GGC	GGA	GGG
His	CAU	CAC		
Leu	CUU	CUC	CUA	CUG
Pro	CCU	CCC	CCA	CCG
Val	GUU	GUC	GUA	GUG

Give the amino acid sequence encoded by mRNA sequence in (b). (1 mark)

(d) ***Not*** all gene mutations will result in a change in phenotype. Give ***two*** reasons why. (2 marks)

##

(a) Substitution (of base) (1)

(b) CCUGUGGAG (1)

(c) Pro – Val – Glu (1)

(d) Any ***two*** of the following: (2)

- The triplet code is degenerate. / Substitution of a base may still result in a codon specifying the same amino acid.
- The mutation may occur in a non-coding region of DNA, which does not code for proteins.
- The change in amino acid sequence does not change the shape of the protein sufficiently.
- The mutant allele is recessive to the normal allele and is not expressed in the heterozygous condition.

(5 marks)

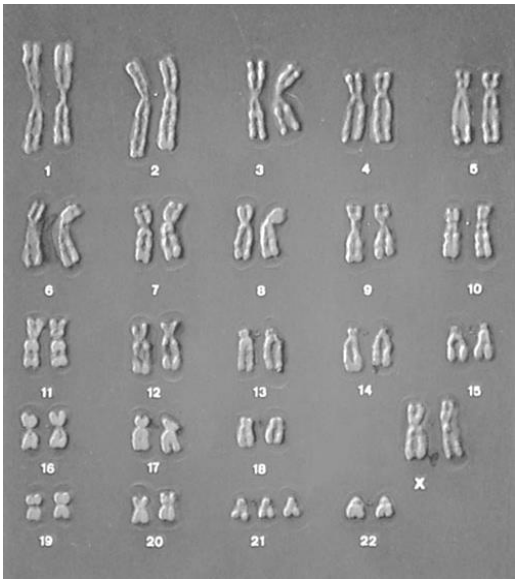
##

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	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
(c)	Analysing data	1
(d)	Applying concepts	2

|!|BCESB0270000003|!| ★ **CRO**

The photomicrograph below shows the karyotype of a foetus with a chromosomal abnormality.



(a) What abnormality is shown in the above karyotype? (1 mark)

_____

(b) What is the sex of the foetus? (1 mark)

_____

(c) Name a procedure by which foetal cells can be obtained to produce the karyotype. (1 mark)

_____

(d) Give ***two*** circumstances under which a doctor would advise pregnant women to have karyotype analysis of the foetus. (2 marks)

_____

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(e) From the karyotype, can we deduce whether this foetus will be red-green colour blind or not? Explain your answer. (2 marks)

##

(a) There is an extra copy of chromosome 21 / Trisomy 21 (1)

(b) Female (1)

(c) Amniocentesis / Chorionic villus sampling (1)

(d) Advanced maternal age (1)
Family history of genetic diseases due to chromosomal abnormalities (1)
(Accept other reasonable answer.)

(e) No, this is because
the karyotype cannot reveal the genetic make-up of the foetus / the karyotype only shows the number or appearance of chromosomes (1)
whereas red-green colour blindness is due to the presence of a certain allele. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	2

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|!|BCESC0270000004|!| **CRO**

DNA replication and transcription are two important processes in cells.

(a) The table below contains statements about DNA replication and transcription. Put a '✓' in the box if the statement is correct, or a '×' if the statement is incorrect.

(2 marks)

Statement	DNA replication	Transcription
Occurs in the nucleus		
DNA unwinds		
Involves complementary base pairing		
Involves RNA polymerase		

(b) Explain the significance of each process. (4 marks)

##

(a)

Statement	DNA replication	Transcription
Occurs in the nucleus	✓	✓
DNA unwinds	✓	✓
Involves complementary base pairing	✓	✓
Involves RNA polymerase	×	✓

1 mark for each column correct (1, 1)

(b) DNA replication produces two identical DNA molecules (1)
so that the same genetic information can by passed from a cell to its daughter cells at cell division / from generation to generation. (1)
During transcription, genetic information in the DNA is copied to mRNA (1)
which serves as a template for protein synthesis. (1)

(6 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	4

|!|BCESC0270000005|!|
The base sequence of the template strand of DNA coding for a segment of polypeptide is shown below.

CCGGGCGCAGGCACG

(a) Give the base sequence of mRNA produced from this DNA template. (1 mark)

(b) Describe the process by which the mRNA obtained direct the synthesis of the polypeptide. (5 marks)

##

(a) GGCCCGCGUCCGUGC (1)

(b) The mRNA move out of the nucleus to the cytoplasm (1)
and binds to a ribosome. (1)
Specific amino acids are carried to the ribosome by tRNA molecules. (1)
The anticodons on the tRNA molecules bind to the complementary codons on the mRNA. (1)
Adjacent amino acids are joined by a peptide bond, forming the polypeptide. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	5

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|!|BCESB0270000006|!| ★ **CRO**

(a) State ***two*** differences between the cells produced by mitosis and meiosis. (2 marks)

(b) Mutations can occur during mitosis and meiosis.

(i) Give ***two*** factors that can increase the rate that mutations occur. (2 marks)

(ii) The table below shows the DNA triplet codes for three different amino acids.

Amino acid	DNA triplet code
Glycine	CCA CCG CCT CCC
Isoleucine	TAA TAG TAT
Methionine	TAC

Some substitution mutations would alter the amino acid sequence of a polypeptide, while others would not. Use the information in the table to explain why. (2 marks)

##

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(a) Any ***two*** of the following: (2)

- Mitosis produces two cells while meiosis produces four cells.
- Mitosis produces body cells / somatic cells while meiosis produces sex cells / gametes.
- Mitosis produces diploid cells while meiosis produces haploid cells.
- Mitosis produces genetic identical cells while meiosis produces genetically different cells.

(b) (i) Exposure to chemical antigens / nitrous acid / tar / dioxin / asbestos / mustard gas (1)

Exposure to high-energy radiations / ionizing radiations / X-rays / gamma rays / ultraviolet light (1)

(ii) The substitution of the third base (base C) in the triplet code TAC will give a different amino acid. (1)

The substitution of the third base in the triplet codes for Glycine will still give the same amino acid. (1)

(Accept other reasonable answers.)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Applying concepts	2

|!|BCESB0270000007|!| ★

Part of the base sequence on the coding strand of DNA of a normal allele and a mutant allele are shown below.

Normal allele: AAGGAAACT
Mutant allele: AAGGTAACT

(a) Give the base sequence on the DNA template of the normal allele. (1 mark)

_____

(b) Give the base sequence of the mRNA for the normal allele. (1 mark)

_____

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The table below shows the 64 codons and the corresponding amino acids and signals they specify.

First base	Second base				Third base
	U	C	A	G	
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	(Stop)	(Stop)	A
	Leu	Ser	(Stop)	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met (Start)	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

(c) Using the table above to determine the amino acid sequence encoded by the normal allele. (1 mark)

_____

(d) A tRNA molecule which of the following anticodons is involved in the synthesis of the corresponding segment of the polypeptide produced from the mutant allele but not in that produced from the normal allele? Put a '✓' in the appropriate box below. (1 mark)

☐ GAA ☐ GUA ☐ CUU ☐ CAU

##

(a) TTCCTTTGA (1)

(b) AAGGAAACU (1)

(c) Lys – Glu – Thr (1)

(d) CAU (1)

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(4 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	1
(c)	Analysing data	1
(d)	Analysing data	1

|!|BCESC0270000008|!|

Complete the following paragraph with appropriate words. (6 marks)

Protein synthesis involves two stages. The first stage is (a)_____ and takes place in the nucleus. During this stage, (b)_____ RNA is made using a DNA strand as a template. The second stage, known as (c)_____, takes place in the cytoplasm at structures called (d)_____. During this stage, (e)_____ RNA molecules with anticodons complementary to the codons on the mRNA bring in amino acids in the correct order. The amino acids are joined together by (f)_____ bonds to form a polypeptide chain.

##

(a) transcription (1)
(b) messenger (1)
(c) translation (1)
(d) ribosomes (1)
(e) transfer (1)
(f) peptide (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

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|!|BCESC0270000009|!| **CRO**

(a) State ***two*** chemical differences between DNA and RNA molecules. (2 marks)

(b) Explain ***one*** way in which the structures of DNA and RNA are suited to their functions. (4 marks)

##

(a) DNA contains thymine while RNA contains uracil instead. (1)
DNA contains deoxyribose while RNA contains ribose. (1)

(b) DNA (any ***one set*** of the following): (2)

- DNA is a large molecule made up of a huge number of nucleotides. (1)
  It can store a very large amount of genetic information. (1)
- DNA is double-stranded with strong sugar-phosphate backbones. (1)
  This makes DNA a very stable molecule for long-time storage. (1)
- Complementary base pairing enables DNA to replicate (1)
  so that genetic information is passed on to new cells. (1)

RNA:

- RNA is a small enough to pass through nuclear pores (1)
  so it carries genetic information to the cytoplasm for translation. (1)

(Accept other reasonable answers.)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	4

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